

Exercise 56

Use the spectral decomposition to show that $K \equiv -i \log(U)$ is Hermitian for any Unitary U , and thus $U = \exp(iK)$ for some Hermitian K .

Suffices to show that $-i \log(U) = (-i \log(U))^\dagger$

Suffices to show that $-i \log(U) = i(\log(U))^\dagger$

Suffices to show that $-i \log(U) = i(\log(U^\dagger))$

Suffices to show that $i \log(U^{-1}) = i \log(U^\dagger)$

Suffices to show that $U^{-1} = U^\dagger$, but unitary operation is defined so. ■

Exercise 57

To show that cascaded measurements are single measurements, we would have to show two things, that the cascaded measurement takes any arbitrary state to the state state as the single measurement and tha the the probability distribution of eigenvalues of the cascaded measurement is the same as that of the single measurement:

i) same eigenvector: Let $|\psi\rangle$ be the initial arbitrary state

post first measurement: $\frac{L_I |\psi\rangle}{\sqrt{\langle\psi|L_I^\dagger L_I|\psi\rangle}}$

post second measurement:

numerator:

$$M_m \left(\frac{L_I |\psi\rangle}{\sqrt{\langle\psi|L_I^\dagger L_I|\psi\rangle}} \right)$$

denominator:

$$= \sqrt{\frac{\langle\psi|L_I^\dagger M_m^\dagger M_m L_I|\psi\rangle}{\langle\psi|L_I^\dagger L_I|\psi\rangle}} = \sqrt{\frac{\langle\psi|L_I^\dagger M_m^\dagger M_m L_I|\psi\rangle}{\langle\psi|L_I^\dagger L_I|\psi\rangle}}$$

the state after second measurement:

$$\begin{aligned} & \frac{M_m L_I |\psi\rangle}{\sqrt{\langle\psi|L_I^\dagger L_I|\psi\rangle} \sqrt{\langle\psi|L_I^\dagger M_m^\dagger M_m L_I|\psi\rangle}} \\ &= \frac{M_m L_I |\psi\rangle}{\sqrt{\langle\psi|L_I^\dagger L_I|\psi\rangle} \sqrt{\langle\psi|L_I^\dagger M_m^\dagger M_m L_I|\psi\rangle}} \\ &= \frac{M_m L_I |\psi\rangle}{\sqrt{\langle\psi|L_I^\dagger M_m^\dagger M_m L_I|\psi\rangle}} \quad \blacksquare \end{aligned}$$

Exercise 58

Suppose we prepare a quantum system in an eigenstate $|\psi\rangle$ of some observable M , with corresponding eigenvalue m . What is the average observed value of M , and the standard deviation?

Average value = $\langle\psi|M|\psi\rangle$

$$\begin{aligned} &\Rightarrow \frac{\langle\psi|M_m^\dagger}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}} M \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}} \\ &= \frac{\langle\psi|M_m^\dagger M M_m|\psi\rangle}{\langle\psi|M_m^\dagger M_m|\psi\rangle} \\ &= \frac{\langle\psi|M_m^\dagger M_m|\psi\rangle}{\langle\psi|M_m^\dagger M_m|\psi\rangle} = 1? \end{aligned}$$

Note that since the effect of the M on M_m was leave it unchanged, the same will happen with M^2 , meaning $\sqrt{\langle M^2 \rangle - \langle M \rangle^2} = \sqrt{1 - 1} = 0$ ■

Bonus

Show that $\sqrt{\langle (M - \langle M \rangle)^2 \rangle} = \sqrt{\langle M^2 \rangle - \langle M \rangle^2}$

Ex 2.59

Suppose we have a qubit in the state $|0\rangle$, and we measure the observable X . What is the average value of X ? What is the standard deviation of X ?

$$\langle X \rangle = \langle 0|X|0 \rangle = \langle 0|1 \rangle = 0$$

$$\sqrt{\langle X^2 \rangle - \langle X \rangle^2} = \sqrt{\langle 0|X|0 \rangle - (\langle 0|X|0 \rangle)^2} = \sqrt{1 - 0} = \pm 1 \blacksquare$$

Ex 2.60

i) Show that $\vec{v} \cdot \vec{\sigma}$ has eigenvalues ± 1 .

Solution:

$$\text{Let } \vec{v} \text{ be } \begin{pmatrix} a \\ b \\ \sqrt{1-a^2-b^2} \end{pmatrix}.$$

$$\Rightarrow \vec{v} \cdot \vec{\sigma} = a\sigma_1 + b\sigma_2 + c\sigma_3$$

$$= \begin{bmatrix} \sqrt{1-a^2-b^2} & a-ib \\ a+ib & -\sqrt{1-a^2-b^2} \end{bmatrix}$$

$$\Rightarrow \text{the characteristic equation is } (\sqrt{1-a^2-b^2} - \lambda)(-\sqrt{1-a^2-b^2} - \lambda) - (a-ib)(a+ib) = 0$$

$$\Rightarrow \lambda^2 - (1-a^2-b^2) - a^2 - b^2 = 0$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1 \quad \blacksquare$$

ii) Show that the projectors onto the corresponding eigenspaces are given by $P_{\pm} = (I \pm \vec{v} \cdot \vec{\sigma})/2$

So long as $\vec{v} \cdot \vec{\sigma}$ defines a measurement,

$$\vec{v} \cdot \vec{\sigma} = \sum_i \lambda_i P_i \text{ (where } P_i \text{ are the projectors.)}$$

$$= (+1P_+) + (-1P_-)$$

$$\Rightarrow \vec{v} \cdot \vec{\sigma} = P_+ - P_- \text{ (Let this be eq 1)}$$

$$I = \sum_i P_i \text{ (Let this be eq 2)}$$

$$\text{Using eq2 in eq 1 we have } \vec{v} \cdot \vec{\sigma} = P_+ - (I + P_+)$$

$$\Rightarrow \vec{v} \cdot \vec{\sigma} = 2P_+ - I \Rightarrow P_+ = (I + \vec{v} \cdot \vec{\sigma})/2$$

Similarly,

$$\text{Using eq2 in eq 1 we have } \vec{v} \cdot \vec{\sigma} = P_+ - (I + P_+)$$

$$\Rightarrow \vec{v} \cdot \vec{\sigma} = I - 2P_- \Rightarrow P_- = (I - \vec{v} \cdot \vec{\sigma})/2$$

$$\text{Thus, } P_{\pm} = (I \pm \vec{v} \cdot \vec{\sigma})/2 \quad \blacksquare$$

Ex 2.61

Calculate the probability of obtaining the result $+1$ for a measurement of $\vec{v} \cdot \vec{\sigma}$, given that the state prior to measurement is $|0\rangle$. What is the state of the system after the measurement if $+1$ is obtained?

Probability of measuring $+1$ when the system is in the state $|0\rangle$ is given by

$$\begin{aligned} p(+1) &= \langle 0 | (\vec{v} \cdot \vec{\sigma}) | 0 \rangle \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{bmatrix} \sqrt{1-a^2-b^2} & a-ib \\ a+ib & -\sqrt{1-a^2-b^2} \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{bmatrix} \sqrt{1-a^2-b^2} \\ a+ib \end{bmatrix} \\ &= \sqrt{1-a^2-b^2} \end{aligned}$$

The state of the system after the measurement is obtained is P_+ ie, $(I + \vec{v} \cdot \vec{\sigma})/2$ ■

Ex 2.62

Show that any measurement where the Measurement operators and the POVM elements coincide is a projective measurement.

POVM elements: P_i such that $\sum_i P_i = I$

Measurement Operators: M_m describe a measurement such that $\langle \psi | M_m | \psi \rangle = pr(m)$

Show that if these two coincide such that we have $\sum_m M_m = I$, then

$M_m M_n = 0$, where $m \neq n$

Solution:

Consider two projectors M_m, M_n such that they are together complete

$$\Rightarrow M_m = I - M_n$$

Now consider $M_m M_n | \psi \rangle$:

$$\begin{aligned} &= M_m (M_n | \psi \rangle) \\ &= (I - M_n) (M_n | \psi \rangle) \\ &= (M_n | \psi \rangle) - (M_n^2 | \psi \rangle) \\ &= M_n | \psi \rangle - M_n | \psi \rangle \\ &= 0 \end{aligned}$$

Thus, $M_m M_n = 0$

This solves it for the case where there are two projectors

What about the case where there are 3 or n projectors? Is it that as long as a projector is orthogonal with its complement with Identity, every pair of projectors conceivable for that observable is orthogonal?

This solution is not right

Ex 2.63

Suppose a measurement is described by measurement operators M_m . Show that there exist unitary operators U_m such that $M_m = U_m \sqrt{E_m}$, where E_m is the POVM associated with the measurement.

Solution: We know that any linear operator can be written as the product of a unitary operator and a positive operator.

Note that E_m is a positive operator. Note also that $\sqrt{E_m}$ is also a positive operator.

Thus, there exists a unitary operator U_m such that $M_m = U_m \sqrt{E_m}$

■